



JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER & SYSTEMS SCIENCES
NEW DELHI-110067

Marks – 30 Time-60 Minutes

Class: MCA. III Semester

Course Title: Wireless Networks

Course Code: CS-446

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- Q1. High-frequency X-rays and Gamma rays are not normally used for wireless communication. Explain why. **[Marks-5]**
- Q2. What is multipath propagation? Explain how it affects signal quality. **[Marks-5]**
- Q3. What is the maximum data rate that can be supported on a 10 MHz noiseless channel, if the channel uses eight-level digital signals? **[Marks-5]**
- Q4. Write the difference among 1G, 2G, 3G, 4G and 5G. **[Marks-5]**
- Q5. A pure ALOHA network transmits 200 – bit frames on a shared channel of 200 kbps. What is the requirement to make this free collision free? **[Marks-5]**
- Q6. Write the classification of propagation models and explain following models in detail, with their applicability. **[Marks-5]**
- a) Free Space model
 - b) Two Ray model



Marks – 70 Time-180 Minutes

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- Q1.Explain the role of interface frame spacing in wireless network. [Marks-5]
- Q2. Write the differences between wireless networks and wired network? [Marks-5]
- Q3. What is the fading and describes its type in detail? [Marks-5]
- Q4. Explain radio propagation mechanism and its types. [Marks-5]
- Q5. Define medium access control in wireless network and how it is different from wired network. [Marks-5]
- Q6. Define Access point (AP), Extended service set (ESS) and Basic service set (BSS). [Marks-5]
- Q7. Derive the Throughput of Pure and Slotted Aloha. [Marks-5]
- Q8. What is Doppler effect and explain its impact on wireless communication? [Marks-5]
- Q9. Explain the Free Space and Two Ray Model and also compare their characteristics. [Marks-10]
- Q10. Explain the IEEE 802.11 Architecture with its functions in detail. [Marks-10]
- Q11. Write the difference among the following. [Marks- 10]
- a) Pure Aloha and Slotted Aloha.
 - b) Non-persistent, one- persistent and p- persistent.